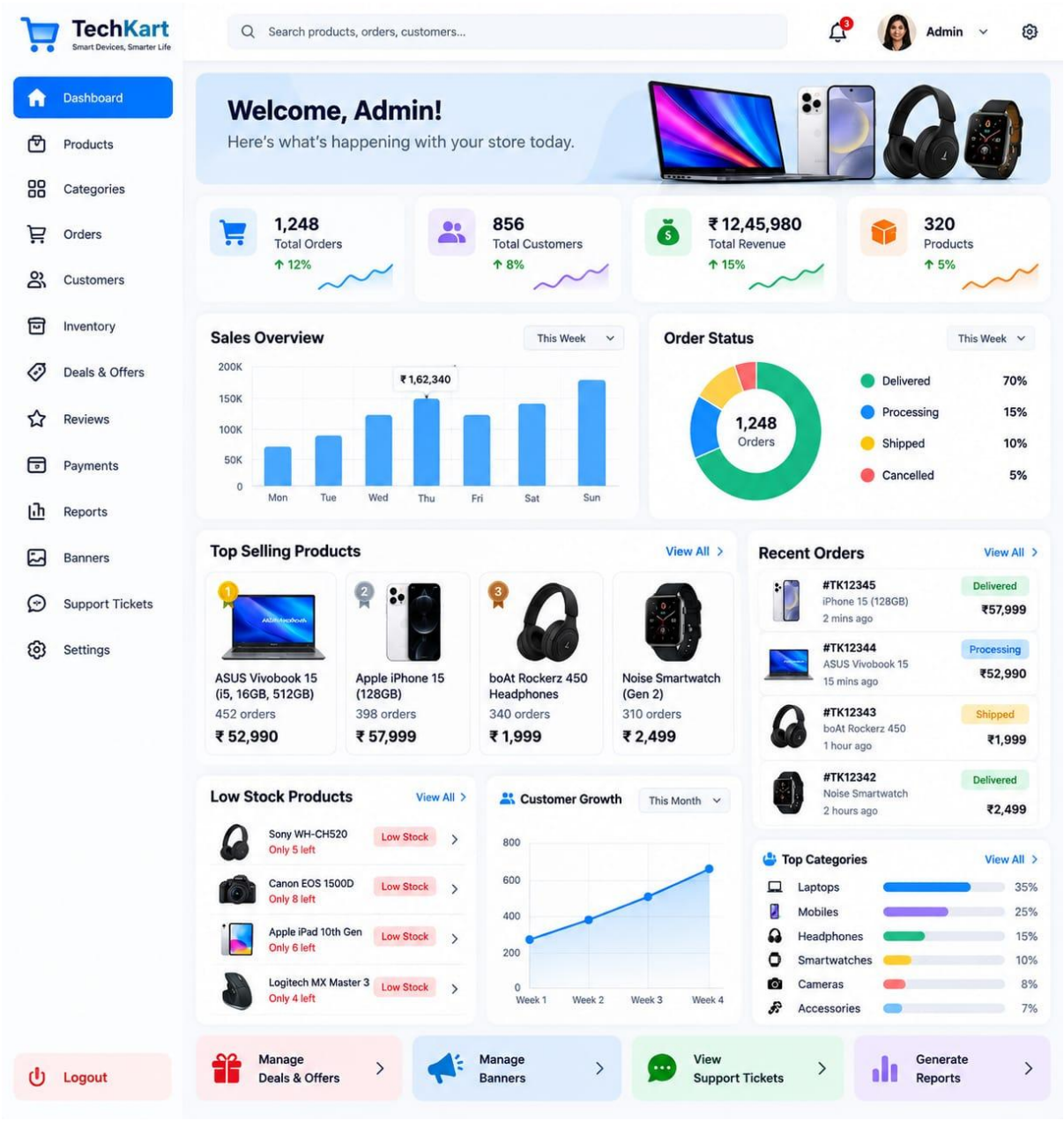
AI-Powered Multi-Domain Commerce and Relationship Management Platform

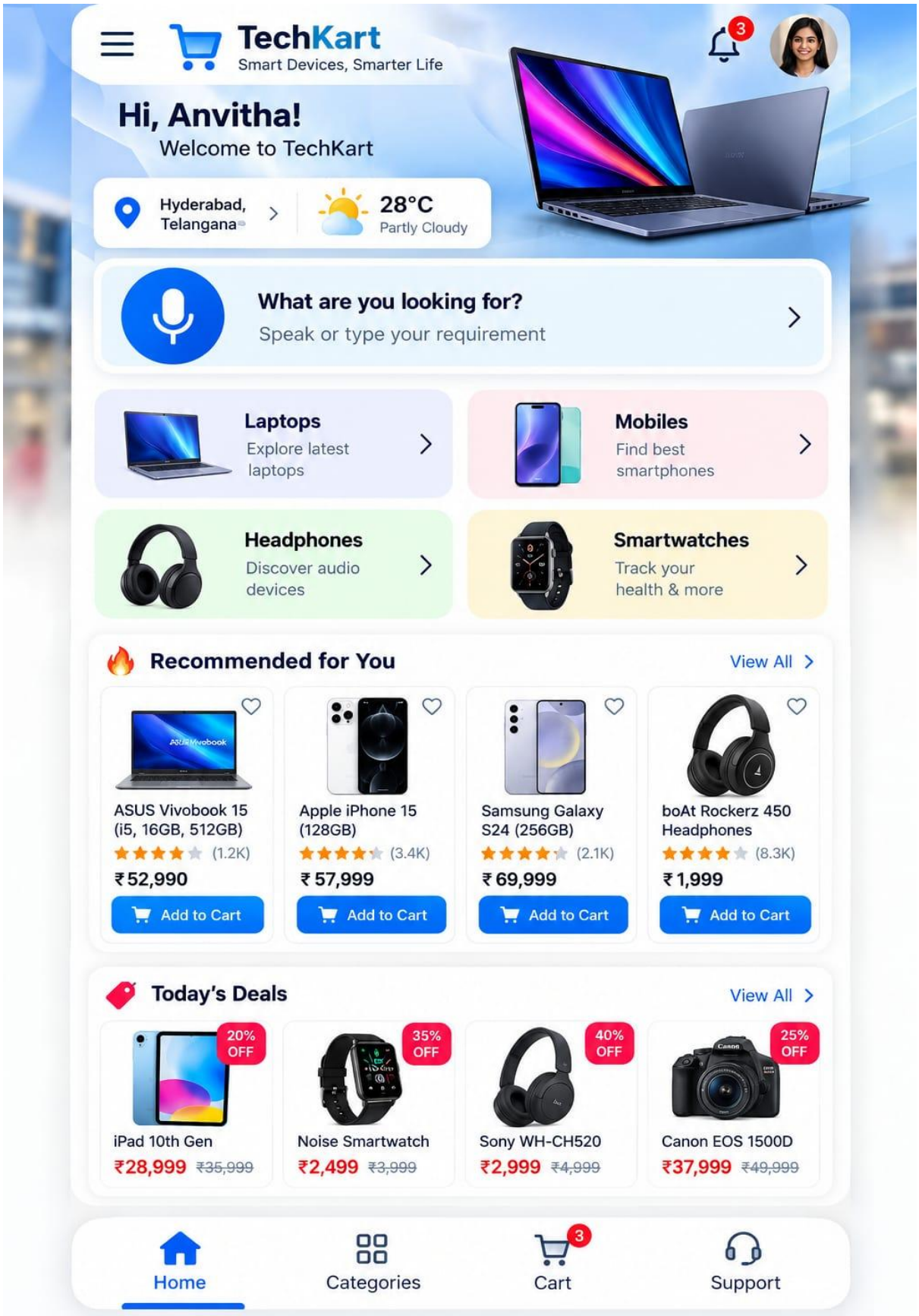
Problem Statement

My project is an AI-Powered Multi-Domain Commerce and Relationship Management Platform that combines E-Commerce and Agriculture into a single intelligent platform. The system provides separate experiences for customers and farmers while using a common AI and relationship-management layer. Users can log in, select their preferred language (English, Telugu, or Hindi), and choose either the E-Commerce or Agriculture domain. In the E-Commerce domain, users can browse and purchase electronic products such as laptops, mobiles, tablets, earphones, smartwatches, cameras, and accessories. In the Agriculture domain, farmers can purchase seeds, fertilizers, and farming instruments and manage their orders and support requests. The platform provides multilingual support, a dynamic voice assistant, and an AI assistant based on RAG (Retrieval-Augmented Generation), a domain-specific Knowledge Base, and a personalized AI agent. The voice assistant converts the user's speech into text, identifies the language and intent, retrieves relevant information from the user's personal data and the appropriate domain knowledge base, and generates a dynamic response that can also be converted back into speech. The personalized AI agent uses each user's profile, search history, viewed products, wishlist, cart, orders, reviews, and previous conversations to provide relevant responses and recommendations without mixing information between users. For new users, the system can initially recommend popular or top-selling products, while recommendations become personalized as user activity increases. The platform can also use sentiment analysis to understand whether a customer or farmer is satisfied, dissatisfied, frustrated, or positive, and a next-best-action mechanism can determine an appropriate response such as providing order information, recommending a product, creating a support request, or escalating an issue. All user interactions, conversations, orders, and activities are stored securely and can be analyzed by the admin. The Admin dashboard manages users, products, orders, inventory, support requests, conversations, AI knowledge bases, and analytics for both domains. Overall, the project combines multilingual AI, voice interaction, RAG, knowledge bases, personalized AI agents, sentiment analysis, recommendations, and relationship management to provide an intelligent and personalized commerce and farmer-support experience.

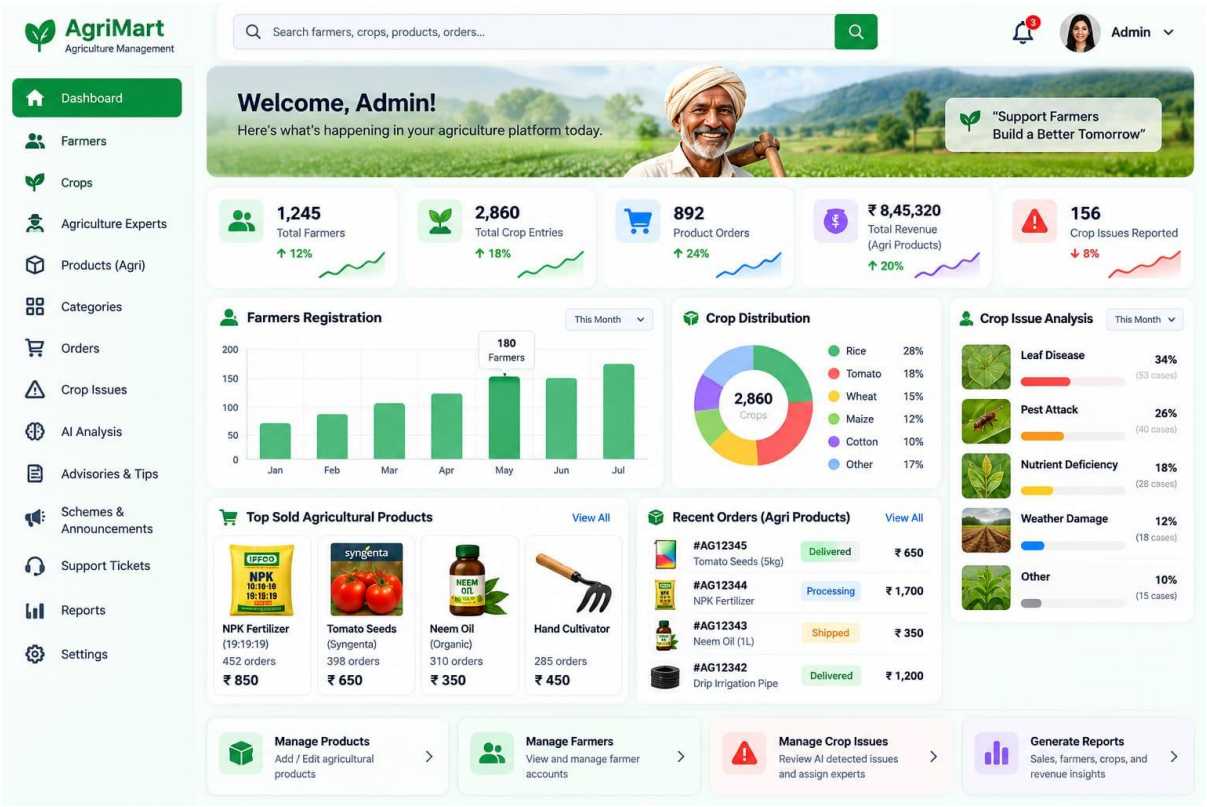
E-Commerce Admin



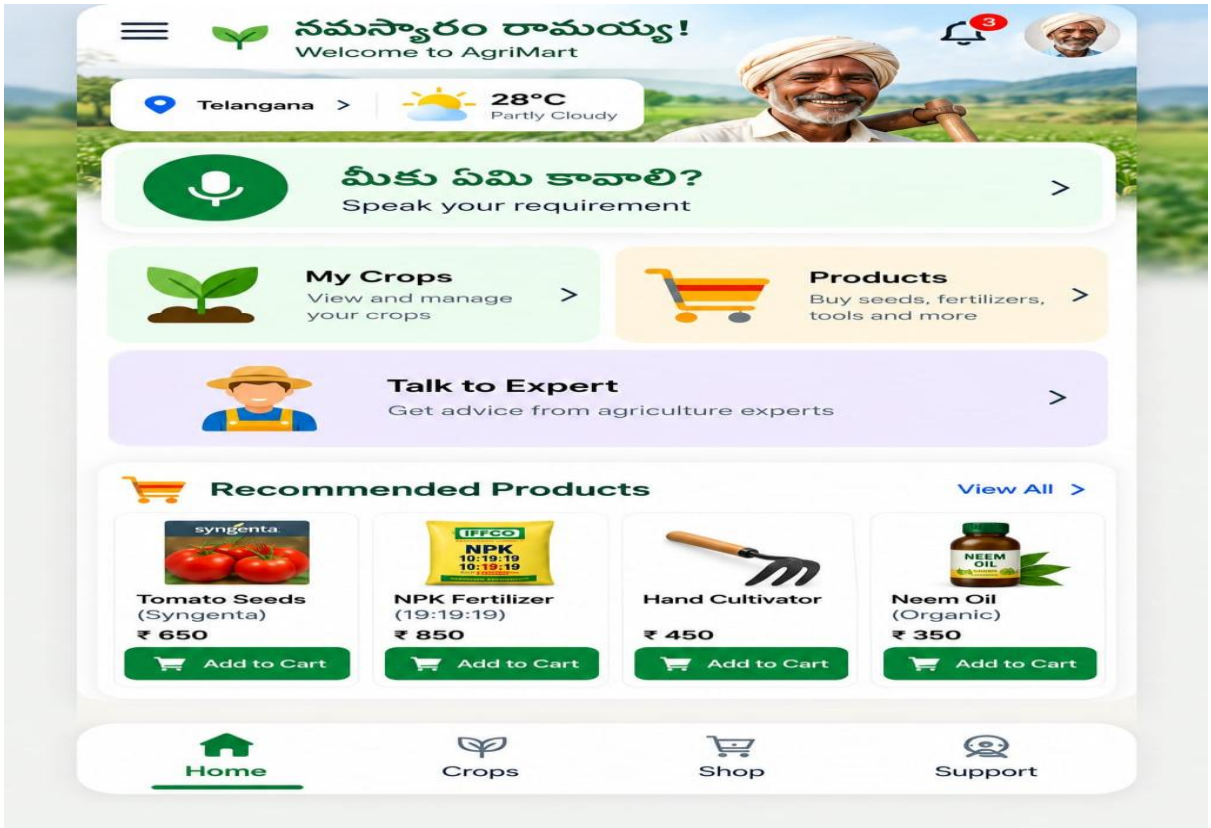
E-Commerce Customer



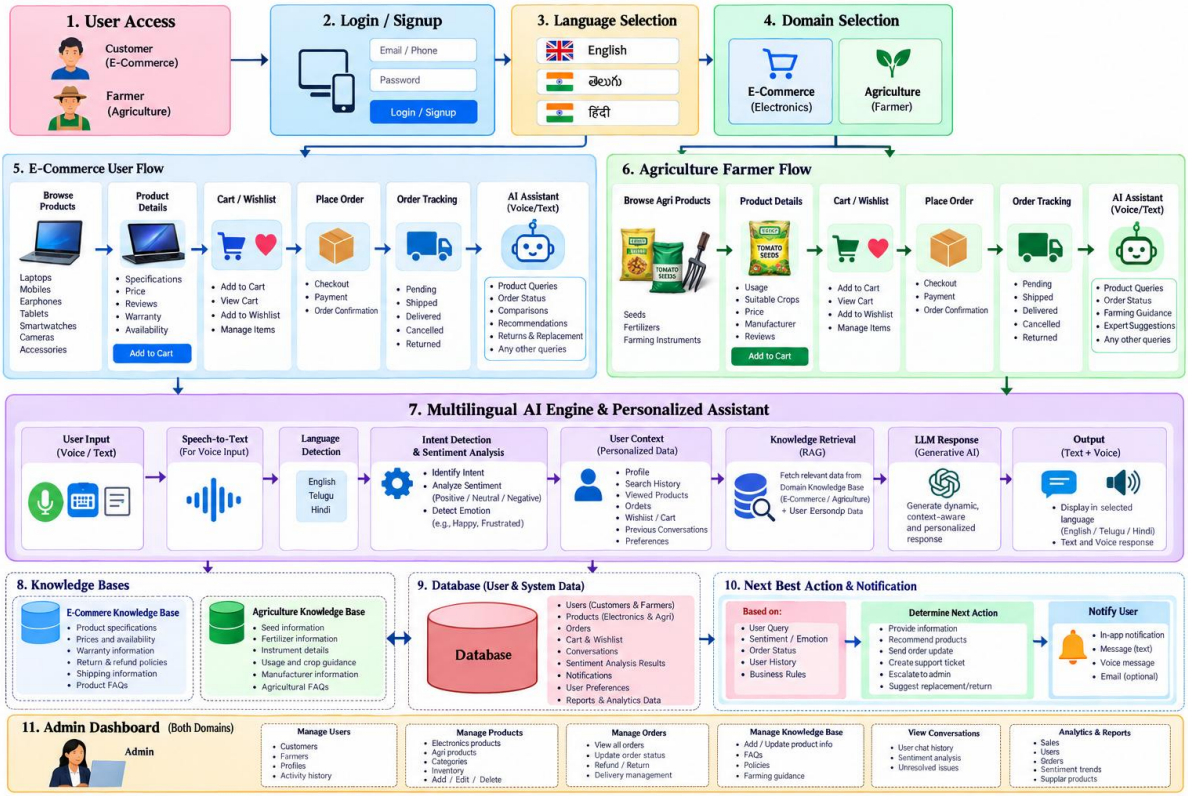
Agriculture Admin



Agriculture Customer



Flow


1. User Access
Customer (E-Commerce)
Farmer (Agriculture)
2. Login / Signup
Email / Phone
Password
Login / Signup
3. Language Selection
English
తెలుగు
हिंदी
4. Domain Selection
E-Commerce (Electronics)
Agriculture (Farmer)
5. E-Commerce User Flow
Browse Products
Laptops
Mobiles
Earphones
Tablets
Smartwatches
Cameras
Accessories
Product Details
Specifications
Price
Reviews
Warranty
Availability
Add to Cart
Cart / Wishlist
Add to Cart
View Cart
Add to Wishlist
Manage Items
Place Order
Checkout
Payment
Order Confirmation
Order Tracking
Pending
Shipped
Delivered
Cancelled
Returned
AI Assistant (Voice/Text)
Product Queries
Order Status
Comparisons
Recommendations
Returns & Replacement
Any other queries
6. Agriculture Farmer Flow
Browse Agri Products
Seeds
Fertilizers
Farming Instruments
Product Details
Usage
Suitable Crops
Price
Manufacturer
Reviews
Add to Cart
Cart / Wishlist
Add to Cart
View Cart
Add to Wishlist
Manage Items
Place Order
Checkout
Payment
Order Confirmation
Order Tracking
Pending
Shipped
Delivered
Cancelled
Returned
AI Assistant (Voice/Text)
Product Queries
Order Status
Farming Guidance
Expert Suggestions
Any other queries
7. Multilingual AI Engine & Personalized Assistant
User Input (Voice / Text)
Speech-to-Text (For Voice Input)
Language Detection
English
Telugu
Hindi
Intent Detection & Sentiment Analysis
Identify Intent
Analyze Sentiment (Positive / Neutral / Negative)
Detect Emotion (e.g., Happy, Frustrated)
User Context (Personalized Data)
Profile
Search History
Viewed Products
Orders
Wishlist / Cart
Previous Conversations
Preferences
Knowledge Retrieval (RAG)
Fetch relevant data from Domain Knowledge Base (E-Commerce / Agriculture)
+ User Eersondp Data
LLM Response (Generative AI)
Generate dynamic, context-aware and personalized response
Output (Text + Voice)
Display in selected language (English / Telugu / Hindi)
Text and Voice response
8. Knowledge Bases
E-Commere Knowledge Base
Product specifications
Prices and availability
Warranty information
Return & refund policies
Shipping information
Product FAQs
Agriculture Knowledge Base
Seed information
Fertilizer information
Instrument details
Usage and crop guidance
Manufacturer information
Agricultural FAQs
9. Database (User & System Data)
Database
Users (Customers & Farmers)
Products (Electronics & Agri)
Orders
Cart & Wishlist
Conversations
Sentiment Analysis Results
Notifications
User Preferences
Reports & Analytics Data
10. Next Best Action & Notification
Based on:
User Query
Sentiment / Emotion
Order Status
User History
Business Rules
Determine Next Action
Provide information
Recommend products
Send order update
Create support ticket
Escalate to admin
Suggest replacement/return
Notify User
In-app notification
Message (text)
Voice message
Email (optional)
11. Admin Dashboard (Both Domains)
Admin
Manage Users
Customers
Farmers
Profiles
Activity history
Manage Products
Electronics products
Agri products
Categories
Inventory
Add / Edit / Delete
Manage Orders
View all orders
Update order status
Refund / Return
Delivery management
Manage Knowledge Base
Add / Update product info
FAQs
Policies
Farming guidance
View Conversations
User chat history
Sentiment analysis
Unresolved issues
Analytics & Reports
Sales
Users
Orders
Sentiment trends
Supplar products